

TUSHAR JAIN

Computer Science & Business Systems Student | Full Stack & Systems Builder

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SUMMARY

3rd-year Computer Science & Business Systems engineering student at BMS College of Engineering (CGPA: 9.12/10) with a focus on building and deploying software that solves practical, everyday problems. Built and shipped multiple production applications, including an offline-first POS platform for local vendors, metro navigation apps with 1,500+ Play Store downloads, and an academic resource portal with 50K+ search impressions. Experienced across full-stack web (React, Next.js, TypeScript), offline-first architectures (IndexedDB, Dexie.js), relational database design (PostgreSQL, Supabase), and embedded systems (ESP32, Arduino).

EDUCATION

BMS College of Engineering (BMSCE) - Bengaluru, Karnataka

2024 - 2028

Bachelor of Engineering - Computer Science & Business Systems (CSBS) | **CGPA: 9.12 / 10.0 (up to 4th Semester)**

EXPERIENCE

PulsePredict (Vital Health Tech) - Co-Founder & Tech Lead

Aug 2025 - Jan 2026

Jaipur, India | Pitched at JIC Accelerator 7.0 | [Project Page](#)

- Prototyped a low-cost IoT wearable combining ESP32, MPU6050 (fall detection), and MAX30102 (heart rate/SpO2) sensors for elderly remote health monitoring.
- Built BLE firmware and a real-time web dashboard transmitting vitals over WebSockets with automated emergency alert triggers.
- Pitched the working hardware and software MVP to investors and mentors at JIC Accelerator 7.0 in Jaipur.

InnoByte Services - Web Developer Intern

Aug 2026 - Sep 2026

Remote | Certificate Ref: IS/A1/R8453 | [Verify Certificate](#)

- Developed responsive web interfaces and frontend components using modern JavaScript, HTML5, and CSS3.
- Collaborated with the engineering team to implement interactive features, resolve production UI bugs, and optimize page load performance across devices.

KEY PROJECTS

1. Billing Pro POS | High-Performance Offline-First POS & Restaurant Management System

[Live Demo](#) | [GitHub](#) | [Project Page](#)

- Built an enterprise-grade, offline-first POS system for restaurants and retail stores that operates reliably without active internet connection using Dexie.js (IndexedDB).
- Implemented sub-10 second billing with dynamic UPI Bharat QR code generation, multi-payment tracking (Cash, Card, UPI, Due), and direct ESC/POS Bluetooth and USB thermal printing.
- Features multi-category inventory, order holds, real-time analytics dashboard, and automated local data export/import.

Tech Stack: React, TypeScript, Vite, Dexie.js (IndexedDB), Tailwind CSS, Web Bluetooth API, ESC/POS Thermal Printing, Vercel

2. Jaipur Ride (Metro) | Public Transit Routing & Fare Engine (1,500+ Downloads)

[Google Play](#) | [Live Demo](#) | [GitHub](#)

- Built an offline-capable public transit app with 1,500+ Google Play downloads, 50,000+ website impressions, and 250+ Monthly Active Users (MAU) in Jaipur, India.
- Implemented a client-side Breadth-First Search (BFS) routing algorithm over a compressed 15KB JSON transit network graph, executing sub-10ms route searches without server calls.
- Provides complete station interchanges, departure schedules, distance calculations, and fare structures.

Tech Stack: React.js, JavaScript, HTML5, CSS3, PWA, Android APK, Vercel

3. NotesCSBS | Academic Document Distribution Platform (10,000+ Impressions)

[Live Platform](#) | [Project Page](#)

- Built a centralized academic resource platform for engineering students, achieving 10,000+ search impressions and 1.17K+ organic clicks with an average Google search position of 4.9.
- Designed PostgreSQL database schema on Supabase with Row Level Security (RLS) for secure student/admin authentication and file metadata routing.
- Integrated Google Drive API with server-side caching to prevent API rate limits during peak exam periods.

Tech Stack: React.js, JavaScript, Supabase (PostgreSQL), SQL, Google Drive API, HTML5, CSS3, Vercel, SEO

4. Namma Ride (Bengaluru Metro) | Multilingual Public Transit App (English, Hindi, Kannada)

[Google Play](#) | [Live Platform](#) | [GitHub](#)

- Developed a lightweight, multilingual Bengaluru Metro journey planner supporting English, Hindi, and Kannada for the city's diverse demographic.
- Engineered lightweight state management in vanilla JavaScript to ensure instant load times and reliable offline navigation on weak cellular networks.
- Published to Google Play Store with 100+ downloads.

Tech Stack: HTML5, CSS3, JavaScript (Vanilla), LocalStorage, Android APK

5. RestaurantOS | Multi-Tenant Cloud Restaurant Management SaaS

[Live Demo](#) | [GitHub](#) | [Project Page](#)

- Engineered a multi-tenant cloud-native SaaS unifying POS, interactive table floor management, reservations, inventory, CRM, and analytics.
- Designed a normalized PostgreSQL schema with 20+ tables, UUID primary keys, Row Level Security (RLS), and PL/pgSQL atomic transaction triggers for instant stock deduction upon billing.
- Implemented live table state synchronization and Kitchen Order Ticket (KOT) routing using Supabase WebSockets.

Tech Stack: React, TypeScript, Vite, Supabase (PostgreSQL), SQL, WebSockets, Vercel

6. Split-Payment System | FinTech Multi-Source Transaction Engine

[GitHub](#) | [LinkedIn Post](#) | [Project Page](#)

- Developed a transactional fintech checkout engine enabling single purchases to be split across multiple payment sources (UPI + Card) simultaneously.
- Designed ACID transaction rollback logic: if any single payment source fails, the system executes an automated rollback to prevent partial deductions or double charges.
- Integrated Razorpay API (test mode) to simulate real-world gateway latencies, retries, and timeout handling.

Tech Stack: JavaScript, Node.js, Express, Razorpay API, REST APIs

7. FlightDeck | Real-Time Flight Tracking & Aviation Telemetry Platform

[Live Platform](#) | [GitHub](#) | [LinkedIn Post](#)

- Built a real-time flight tracking platform providing live arrivals, departures, delays, baggage belt assignments, and destination weather telemetry.
- Implemented Next.js server route handlers with a 60-second TTL caching layer to proxy external AirLabs aviation API requests and prevent rate-limit exhaustion.

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, AirLabs API, REST APIs, Vercel

RESEARCH WORK

1. Human-Supervised Dual-UUV System for Autonomous Maritime Surveillance

[ResearchGate Publication](#) | [Research Paper PDF](#) | [Overview Page](#)

- Proposed a split-architecture dual-UUV surveillance framework combining autonomous acoustic threat detection (passive vehicle) with operator-supervised target inspection (active vehicle).
- Applied Extended Kalman Filters (EKF) for underwater localization under acoustic communication constraints, incorporating a strict Human-in-the-Loop safety protocol for maritime security operations.

Core Areas: [Autonomous Systems](#), [Sensor Fusion](#), [Human-in-the-Loop Control](#), [EKF Localization](#), [Acoustic Comms](#)

2. Statistical Analysis: Does Mobile Detox Improve Sleep Quality?

[Research Report PDF](#) | [Overview Page](#)

- Conducted an experimental study with 22 student participants under the BMSCE Data Science framework to investigate physiological sleep metrics.
- Engineered a weighted Sleep Quality Index (SQI) formula and performed parametric two-sample t-testing ($p < 0.01$, $t = 3.18$, $df = 20$), demonstrating statistically significant reduction in sleep onset latency.

Core Areas: [Statistical Modeling](#), [Hypothesis Testing \(t-Test\)](#), [Feature Engineering \(SQI\)](#), [Data Science](#)

SKILLS & TECHNOLOGIES

Languages: JavaScript (ES6+), TypeScript, C, C++, Python, SQL, HTML5, CSS3

Frontend & Mobile: React.js, Next.js, Vite, Tailwind CSS, Vanilla JS, PWA, Android (APK Build & Publish)

Backend & Storage: Node.js, Express, Supabase (PostgreSQL, RLS), Firebase, Dexie.js (IndexedDB), REST APIs

Hardware & IoT: ESP32, Arduino, MPU6050 (IMU), MAX30102 (SpO2), BLE/Bluetooth, I2C/SPI/UART, Sensor Fusion, C++ Firmware

Tools & Platforms: Git, GitHub, Vercel, Postman, VS Code, Google Search Console, Arduino IDE

ACHIEVEMENTS & HACKATHONS

- **Grand Finalist - HackmatriX 2026:** Selected for and participated in the Grand Finale of the 24-hour national hackathon organized by IEEE Computer Society at MITS Gwalior (Ref: MITS-DU-202608-00357).
- **Top 10 Finalist - Conference 2.0 Hackathon (2026):** Selected among 500+ participating teams for developing Safe Route, an AI-powered safer route planning solution.
- **JIC Accelerator 7.0 (2025):** Pitched PulsePredict, a low-cost embedded health-tech wearable for elderly fall detection and remote vital monitoring.
- **Jaipur Ride Public Adoption:** Achieved 1,500+ Google Play Store downloads, 50,000+ website impressions, and 250+ Monthly Active Users.

CERTIFICATIONS

- **Web Development Internship Certificate - InnoByte Services (2026)** | Ref: IS/A1/R8453 | [Verify Credential](#)
- **Investment Risk Management - Coursera (2025)** | Verify: MWIVK1DVGR6J | [Verify Credential](#)
- **Business Analysis & Process Management - Coursera (2025)** | Verify: 76W901BZ2HPB | [Verify Credential](#)
- **Web Development Internship Certificate - 3Skill Training (2025)** | Ref: INTERN260467 | [Verify Credential](#)